A one-dimensional higher-order closability obstruction

Candidate partial result for arXiv:2311.08058

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Status. Candidate partial result, likely valid, awaiting human review. The result settles the one-dimensional finite-output-exponent subcase of the source question and proves a stronger arbitrary-order statement.

1 Source question

Alberti, Bate, and Marchese study closability of differential operators with respect to Radon measures in [1]. In concluding Remark 1.7(iv), page 4, they point out that their first-order analysis does not readily cover the following second-order expectation.

> does not imply the closability of the gradient operator from $L^q(\mu)$ to $L^p(\mu)$, not even in dimension $d = 1$. For example, let μ be the restriction of the Lebesgue measure $\mathscr{L}^1$ to a totally disconnected compact subset of $\mathbb{R}$; then it is easy to prove that the derivative is not closable from $L^q(\mu)$ to $L^p(\mu)$ for any $1 \leq p, q \leq \infty$.
>
> A precise characterization of the (absolutely continuous) measures μ on $\mathbb{R}$ such that the derivative is closable from $L^2(\mu)$ to $L^2(\mu)$ has been given in [4, Theorem 2.2] (see also [15, Theorem 3.1.6]).
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> (iv) Theorem 1.3 can be easily extended to more general *first order* differential operators than just directional derivatives, thus obtaining a statement that includes Corollaries 1.5 and 1.6 as particular cases (see Remark 3.2).
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> However, it seems that *second order differential operators* are not easily included in our analysis. For instance, one would guess that the Laplace operator (defined on $C^2_c(\mathbb{R}^d)$) should be closable from $L^p(\mu)$ to $L^q(\mu)$ only if μ is absolutely continuous with respect to $\mathscr{L}^d$, but such statement does not seem to follow from any of the results in this paper (at least, not easily).
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> (v) At the core of the proof of Theorem 1.3(ii), and of the "only if" part of Corollaries 1.5 and 1.6, are the so-called "width functions", which are taken from

Transcription. For the Laplace operator on $C^2_c(\mathbb{R}^d)$, one would expect closability from $L^p(\mu)$ to $L^q(\mu)$ only when $\mu \ll \mathcal{L}^d$, but the source's results do not easily imply this.

Here closability is sequential: if $u_n \to 0$ in the input space and $Tu_n \to w$ in the output space, then necessarily $w = 0$.

2 Idea of the proof

In one dimension, compact support can be forced by finitely many moment conditions. If μ has a singular part, choose a compact Lebesgue-null set K of positive μ-mass. A smooth cutoff can equal one on K while having arbitrarily small Lebesgue support and arbitrarily small μ-mass off K. We add tiny fixed bumps away from K to kill the first m moments. The m-fold primitive is then compactly supported and all its lower derivatives are uniformly small, whereas its mth derivative converges in $L^q(\mu)$ to $\mathbf{1}_K$. The leading coefficient of a differential operator therefore survives in the output while the input converges to zero.

3 The result

Theorem 1. *Let μ be a positive Radon measure on $\mathbb{R}$, let $1 \le p \le \infty$ and $1 \le q < \infty$, and let*

$$P(D) = \sum_{j=0}^{m} a_j D^j, \qquad m \ge 1, \quad a_m \ne 0,$$

be a scalar constant-coefficient differential operator. Regard $P(D)$ on $C_c^\infty(\mathbb{R})$ as an operator from $L^p(\mu)$ to $L^q(\mu)$. If $\mu \not\ll \mathcal{L}^1$, then $P(D)$ is not sequentially closable.

Consequently, closability forces $\mu \ll \mathcal{L}^1$. In particular, for $m = 2$ this proves the necessary condition suggested in [1, Remark 1.7(iv)] in dimension one for every finite output exponent.

Proof. Write the Lebesgue decomposition as $\mu = f\mathcal{L}^1 + \mu_s$. Since $\mu \not\ll \mathcal{L}^1$, the singular part is nonzero. By singularity and inner regularity there is a compact set $K \subset \mathbb{R}$ such that

$$\mathcal{L}^1(K) = 0, \qquad 0 < \mu(K) = \mu_s(K) < \infty.$$

Choose a bounded open interval I containing K. Outer regularity of both $\mathcal{L}^1$ and μ, followed by intersection of the two neighborhoods, gives open sets U_n with

$$K \subset U_n \Subset I, \qquad \mathcal{L}^1(U_n) \longrightarrow 0, \qquad \mu(U_n \setminus K) \longrightarrow 0.$$

Choose $g_n \in C_c^\infty(U_n)$ such that $0 \le g_n \le 1$ and $g_n = 1$ on a neighborhood of K. Because I is bounded,

$$b_{n,k} := \int_{\mathbb{R}} x^k g_n(x)\, dx \longrightarrow 0 \quad (0 \le k \le m-1). \tag{1}$$

We next fix the moment correctors. There exist $\psi_0, \dots, \psi_{m-1} \in C_c^\infty(\mathbb{R} \setminus \overline{I})$, all supported in one compact set, such that the matrix

$$M_{k\ell} := \int_{\mathbb{R}} x^k \psi_\ell(x)\, dx, \qquad 0 \le k, \ell \le m-1,$$

is invertible. Indeed, take unit-mass smooth bumps concentrated near m distinct points outside $\overline{I}$; as their radii tend to zero, M tends to the corresponding Vandermonde matrix.

Let $c_n = (c_{n,0}, \dots, c_{n,m-1})$ solve $Mc_n = -b_n$, and set

$$h_n := g_n + \sum_{\ell=0}^{m-1} c_{n,\ell}\psi_\ell.$$

By (1), $c_n \to 0$. Thus

$$\int x^k h_n(x)\, dx = 0 \quad (0 \le k \le m-1), \qquad \|h_n\|_{L^1(dx)} \longrightarrow 0, \tag{2}$$

and all h_n are supported in one fixed compact set.

Define the m-fold one-sided primitive

$$u_n(x) := \frac{1}{(m-1)!} \int_{-\infty}^{x} (x-t)^{m-1} h_n(t)\, dt.$$

Then $u_n^{(m)} = h_n$. To the left of the common support, $u_n = 0$. To the right, expand $(x-t)^{m-1}$ by the binomial theorem; every term contains one of the moments in (2), so $u_n = 0$ there as well. The same calculation applies to all lower derivatives. Hence $u_n \in C_c^\infty(\mathbb{R})$, with support in one fixed compact set C. Moreover, for each $0 \le j < m$,

$$\|u_n^{(j)}\|_\infty \le C_j \|h_n\|_{L^1(dx)} \longrightarrow 0. \tag{3}$$

Since $\mu(C) < \infty$, (3) yields $u_n \to 0$ in $L^p(\mu)$ for every $1 \le p \le \infty$.

On K, the correction bumps vanish and $g_n = 1$. Off K, the cutoff error is supported in $U_n \setminus K$. Therefore, for finite q,

$$\|g_n - \mathbf{1}_K\|_{L^q(\mu)}^q \le \mu(U_n \setminus K) \longrightarrow 0.$$

The correction coefficients tend to zero and the fixed bumps have finite $L^q(\mu)$ norm, so

$$h_n \longrightarrow \mathbf{1}_K \quad \text{in } L^q(\mu). \tag{4}$$

Combining (3) and (4) gives

$$P(D)u_n = a_m h_n + \sum_{j=0}^{m-1} a_j u_n^{(j)} \longrightarrow a_m \mathbf{1}_K \quad \text{in } L^q(\mu).$$

The limit is nonzero because $a_m \neq 0$ and $\mu(K) > 0$. Thus the closure of the graph contains both $(0, 0)$ and $(0, a_m \mathbf{1}_K)$, so it is not a graph. This proves nonclosability. $\square$

4 Verification and limitations

The proof is purely analytic. The moment matrix is a small perturbation of a Vandermonde matrix; the moment identities make the one-sided primitive compactly supported; and all convergence occurs on a fixed compact set of finite μ-mass. These points also cover infinite Radon measures and the input endpoint $p = \infty$.

The restriction $q < \infty$ is essential to this argument: small μ-measure of $U_n \setminus K$ does not imply an $L^\infty(\mu)$ bound. No assertion is made for $d \ge 2$, where compact support of a solution to $\Delta u = h$ is not controlled by finitely many scalar moments. Absolute continuity is proved only to be necessary, not sufficient.

5 Novelty check

The lightweight run indexes were searched for the source id and the terms “second-order closability”, “Laplacian”, and “higher-order constant-coefficient operator”. A bounded arXiv search through 21 July 2026 used exact and close variants of “second derivative closable Radon measure”, “Laplacian closable absolutely continuous measure”, and “higher order differential operator weighted L^p closability”. It found the source paper but no later exact answer. This supports, but does not prove, bibliographic novelty.

6 Human-review recommendation

Review as a likely valid partial theorem. The highest-value check is the moment-cancellation argument that makes every u_n compactly supported. If confirmed, the result gives a complete affirmative answer to the source’s necessary-condition question in dimension one for finite output exponent, and simultaneously covers all positive-order constant-coefficient operators.

References

[1] G. Alberti, D. Bate, and A. Marchese, *On the closability of differential operators*, J. Funct. Anal. 289 (2025), 111029; arXiv:2311.08058v2.